Enhancing Human Summaries for Question-Answer Generation in Education

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Introduction

- Automated generation of question-answer pairs for education can be used to **assist students with self-guided reviews** of educational materials or to **support instructors with the creation of assessment materials**.
- A **key challenge** for these question generation (QG) models is to **ensure relevancy** of generated questions.
- Previous work on this problem showed that **using summaries as input improves the quality of question generation (QG)** over original textbook text and that **human-written summaries result in higher quality** QG than automatic summaries.
- Although human-written summaries are better, we find **wide variation in the quality of summaries**. We propose a new methodology that rewrites lower quality summaries into higher quality summaries.
- We conduct **a large scale human evaluation** of generated question-answer pairs from our method and baselines, with **9.2k human annotations** of generated questions

Methods

- **Data:** We collected **human-written summaries** from a total of **392** undergraduate and master's students enrolled in a graduate-level Artificial Intelligence course. Students wrote summaries of **56 sections** of 14 chapters of the textbook "Artificial Intelligence: A Modern Approach" and **40 sections** of 6 chapters of the textbook "Speech and Language Processing."

Step - Description
1. Zero-shot - We generated new summaries from the human-written bullet style summaries with GPT-3 using the following prompt: *"Here's an outline, please expand it into full sentences and paragraphs: {human-written summary in bullet style with incomplete sentences}"*
2. Few-shot - We reviewed 10 examples by fact-checking and removing repeated phrases. We added these examples to the prompt and then generated 100 more summaries out of the hand-written summaries in bullet style.
3. Fine-tuning - We fine-tuned GPT-3's Davinci model with the 100 summaries generated from the few-shot stage. The format of the fine-tuned model was the following: StudentSummary: <bullet point summary> GPT3Summary: <paragraph style generated summary>

Table 1: Description of bootstrapping process to modify the human-written summary style and quality.

Step - Description
1. Zero-shot - We generated QA pairs with GPT-3 using the prompt *"Write 5 to 10 questions along with their corresponding answers from the summary." + "Summary: " + student_summary + "Question: <Text of question.>+ "Corresponding answer:<Text of corresponding answer.>"*
2. Few-shot - We reviewed 20 examples by fact-checking and formatting. We added these examples to the prompt and then generated QA pairs out of the summaries generated with our summary enhancer fine-tuned model.
3. Fine-tuning - We fine-tuned a model with QA pairs generated from the few-shot stage.

Table 2: Description of bootstrapping process to generate QA pairs from a text.

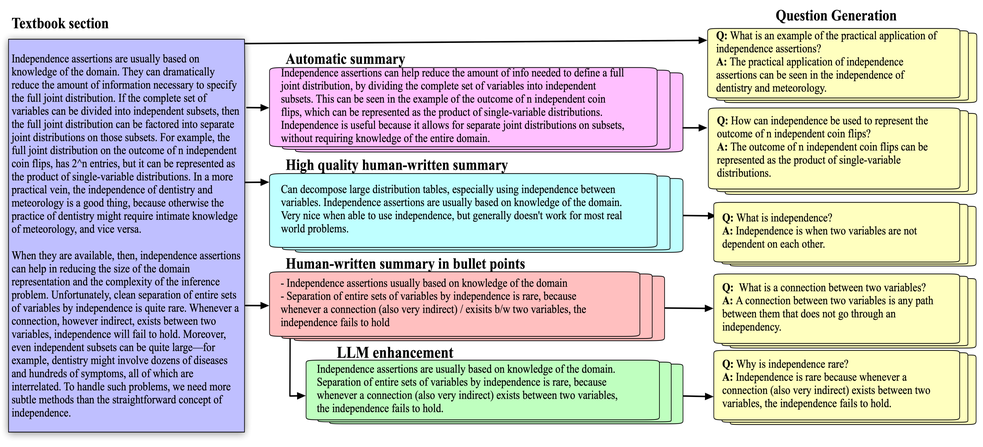

Figure 1: Different types of summaries such as automatic summaries, high quality human-written summaries, poorer quality human-written summary in bullet points, enhanced poorer quality summaries (with our proposed method), and textbook text as used input to QG.

Results

- Human summaries as input to QG models still better than GPT-3 summarizes.
- **But** human-written summaries vary a lot in their own quality
- Enhancing the lower quality summaries, brings them up to the level of usefulness of the higher quality summaries.

Type of Input to QG Model	GPT-3 Acc.	GPT-3 Gram.	GPT-3 Interp.	GPT-3 Rel.	GPT-3 Corr.
1) Original text from textbook	50%	79%	71%	77%	59%
2) GPT-3 generated summary from textbook text	67%	93%	84%	86%	75%
3) Higher quality human-written summary	**73%**	95%	**92%**	**95%**	79%
4) Poorer quality human-written summary	53%	93%	86%	89%	66%
5) Bull2Sum summary	70%	**96%**	90%	93%	**80%**

Table 3: Evaluation of questions generation by GPT-3 using different types of summaries as input.

Future Work

Explore a broader range of course subjects and other types of questions, like multiple choice.

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